
COURSE WORK
PGDS_2025_1 BATCH
MSc Data Science : Coventry University UK

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Index: COMSCDS251P-008

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Executive Summary

The report includes three projects which combine Python development work with data analysis tasks and ethical AI research.

The University Management System operates on Python to demonstrate advanced OOP principles through method overriding and inheritance and polymorphism and encapsulation. The system enables department management and student enrollment and faculty duty assignment and course administration and grading through a scalable framework which includes dynamic course registration and secure record management and GPA calculation.

The E-commerce Data Analysis project involved data cleaning and statistical processing and visualization and predictive modeling after collecting book data from source. The analysis reveals three key findings which show popular categories and stock patterns and pricing and rating behaviors specific to certain product groups and useful pricing recommendations.

The AI Ethics in Healthcare study examines healthcare AI and data science ethical issues through a rapid assessment of bias and privacy and algorithmic fairness while emphasizing responsible AI deployment transparency. The combination of these projects shows how software engineering and data science and ethical frameworks create dependable solutions for business and healthcare and academic institutions.

01. University Management System – Question 01

1. Introduction

There are thousands of products with different prices, categories, and availability on e-commerce portals. Organizations can forecast demand, control stock, and suggest products to customers more efficiently if they know these trends.

In this report, the design and implementation of a University Management System (UMS) have been presented through the aid of Python. It was our intention to incorporate a basic school management system into a more complete platform for various departments, enrollments for courses, grading, academic status tracking, encapsulation, and polymorphism.

Classes are distributed over several files (person.py, student.py, faculty.py, main.py and department.py), and the system is implemented based on modulation. The system is easier to maintain and scale due to this compartmentalization, and it also results in a clean design.

2. Architecture and Design Decisions

Inheritance Hierarchy

The Inheritance structure constitutes the core of the design. A Person base class contains generic properties such as id, full name, and email. This allows shared methods such as `get_responsibilities()` to be inherited and supported.

Person → Student, Faculty, Staff

- Student addresses enrollments, GPA, and academic standing.
- Faculty is indicative of teaching faculty and contains tasks such as teaching and supervision.
- Staff represents non-teaching administrative personnel.

Faculty → Professor, Lecturer, TA

- Each class overwrites `get_responsibilities()` and `calculate_workload()` to allocate different responsibilities and workloads.

Student → UndergraduateStudent, GraduateStudent

- Undergraduates have course work completion as the priority.
- Graduates incorporate research and thesis work into responsibilities.

This hierarchy shows method inheritance and method overriding, with flexibility at the cost of little code duplication.

```
@dataclass
class Person:
    """
    Base class for all people in the system.
    """
    id: Optional[int]
    full_name: str
    email: Optional[str]

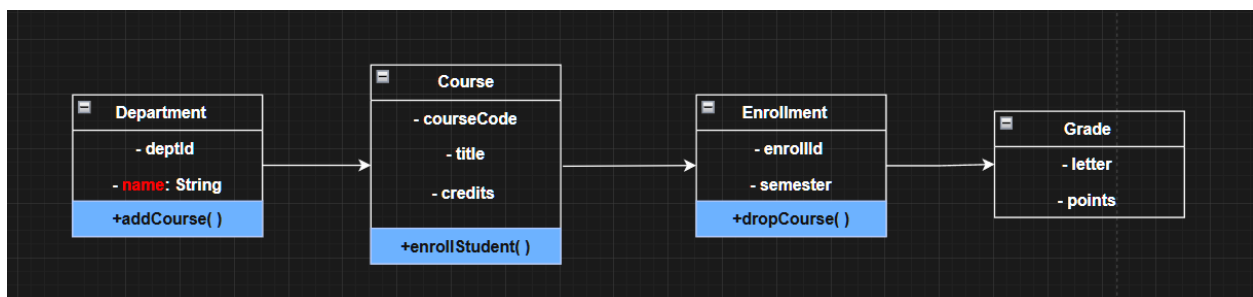
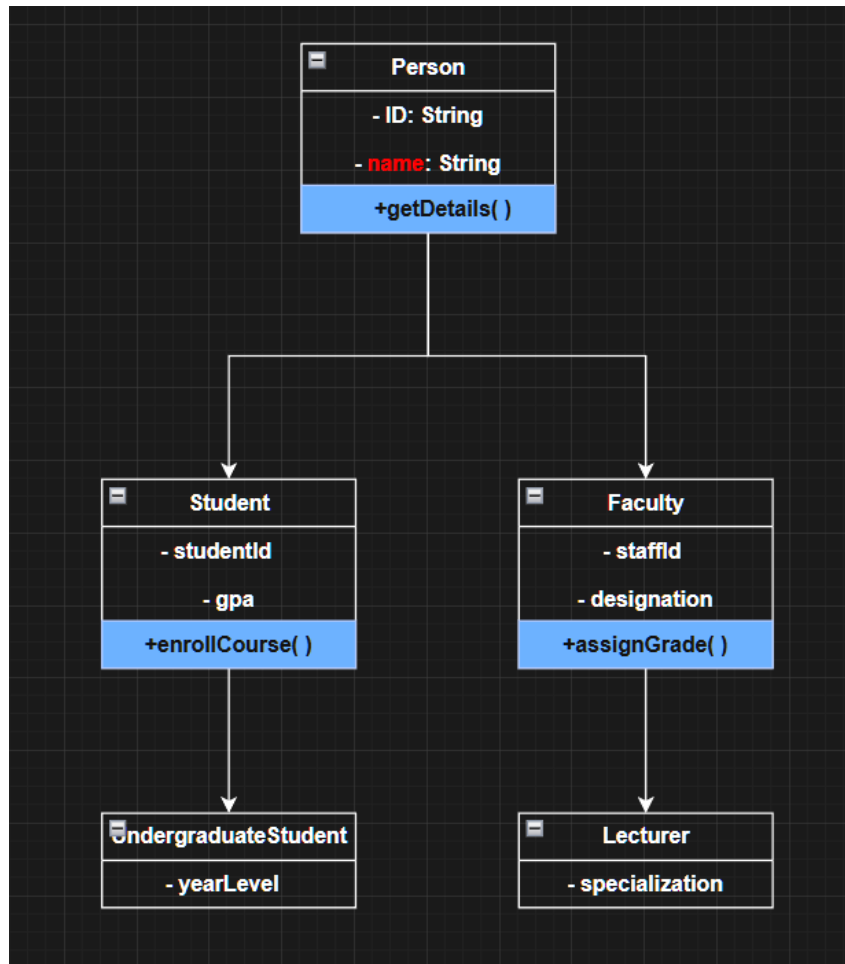
    def get_responsibilities(self) -> str:
        """Generic responsibilities (override in subclasses)."""
        return "General Responsibilities for a person."

    def __repr__(self) -> str:
        return f"<{self.__class__.__name__} name={self.full_name} id={self.id}>"

class Staff(Person):
    """
    Generic Staff member (non-teaching).
    """
    def __init__(self, id: Optional[int], full_name: str, email: Optional[str], position: str):
        super().__init__(id=id, full_name=full_name, email=email)
        self.position = position

    def get_responsibilities(self) -> str:
        return f"Administrative responsibilities as {self.position}."
```

The inheritance tree and connections between the University Management System's core entities are depicted in the UML class diagram that follows.



Advanced Student Management

The student class was enriched with a comprehensive record system for academics:

Course Enrollment

`enroll_course(course_code, credits, semester)` ensures:

- Credit limits are not exceeded (`max_credits_per_semester`).
- Duplicate registration for one and the same course is eliminated.

Dropping Courses

- `drop_course(course_code, semester)` checks whether a course can be dropped. Graded courses can not be dropped to avoid integrity loss.

GPA Calculation

- `calculate_gpa()` adds graded courses by semesters on a 4.0 GPA scale, excluding ungraded records.

Academic Status Tracking

- `get_academic_status()` applies policies:
Dean's List: 12+ credits and a GPA ≥ 3.7
Probation: 6+ credits and a GPA < 2.0
Otherwise: Good Standing

This design enforces real-world academic policies and facilitates extension to other institutions.

Outputs:

```
=== Registration attempts ===
Expected fail (prereq): Prerequisites not satisfied.
Expected fail (capacity): Course capacity reached.

Course summary:
{'course_code': 'CS101', 'name': 'Intro to Programming', 'credits': 3, 'capacity': 3, 'enrolled': 3, 'prereqs': []}
{'course_code': 'CS201', 'name': 'Data Visualization', 'credits': 4, 'capacity': 2, 'enrolled': 2, 'prereqs': ['CS101']}
{'course_code': 'CS301', 'name': 'Machine Learning', 'credits': 4, 'capacity': 2, 'enrolled': 0, 'prereqs': ['CS201']}

GPAs and statuses:
Avishka Jayamanna GPA: 3.829 Status: Good Standing
Nawanjana Pathirage GPA: 3.129 Status: Good Standing
```

```
After drop, CS201 enrolled: ['MSU001']
```

Encapsulation and Validation

To ensure protection of sensitive data, a wrapper `SecureStudentRecord` class was included:

Encapsulation

- Stores private fields such as `_private notes`.
- Provides access control through `@property` getters and setters.

Validation

- Fields GPAs are constrained to 0.0–4.0.
- Notes must be strings with length under 2000 characters.
- Grades must be of the valid grading scheme (A–F).
- This ensures data integrity, prevents invalid states, and simulates secure storage of academic records.

```
@private_notes.setter
def private_notes(self, note: str):
    if not isinstance(note, str):
        raise TypeError("Notes must be text.")
    if len(note) > 2000:
        raise ValueError("Notes too long.")
    self._private_notes = note
```

```
# Grade to points mapping (4.0 scale)
GRADE_POINTS = {
    "A+": 4.0, "A": 4.0, "A-": 3.7,
    "B+": 3.3, "B": 3.0, "B-": 2.7,
    "C+": 2.3, "C": 2.0, "C-": 1.7,
    "D": 1.0, "F": 0.0
}
```


Polymorphism with Method Overriding

The system demonstrates polymorphism in various ways:

Responsibilities:

- Professor.get_responsibilities() → "Lead research, supervise students."
- Lecturer.get_responsibilities() → "Deliver lectures, mark assignments."
- TA.get_responsibilities() → "Assist teaching, run labs."
- Student.get_responsibilities() → "Attend classes."

Workload Calculation

- Professor.calculate_workload() adds research load.
- Lecturer.calculate_workload() uses higher teaching intensity.
- TA.calculate_workload() returns decreased teaching support workload.

When a list of Person objects is traversed, invoking get_responsibilities() calls different implementations depending on the type of object, demonstrating runtime polymorphism.

Outputs:

```
Avishka Jayamanna (UndergraduateStudent): Attend classes, submit assignments, follow program requirements. (Undergraduate responsibilities: complete core modules). workload_metric=N/A
Janith Perera (GraduateStudent): Attend classes, submit assignments, follow program requirements. (Graduate responsibilities: research, thesis). workload_metric=N/A
Dr. Kumara Perera (Professor): Lead research, supervise students, teach advanced courses. workload_metric=1.5
Mr. Janith Silva (Lecturer): Deliver lectures, mark assignments, hold office hours. workload_metric=0.8
Sarah Silva (TA): Assist teaching, run labs, grade under supervision. workload_metric=0.5
```

```
Faculty workloads:
Dr. Kumara Perera (Professor) workload -> 3.5
Mr. Janith Silva (Lecturer) workload -> 2.4
Sarah Silva (TA) workload -> 1.0
```

Department and Course Management

Two additional classes finish the university organization:

Course

- Fields - course_code, name, credits, capacity, prerequisites.
- Manages enrollment capacity and prevents over-capacity.

Department

- Stores lists of courses and faculty.

Methods:

- add_course() – inserts a new course.
- assign_faculty() – assigns instructors to courses.
- enroll() – performs checks on prerequisites and capacity before admitting students.
- drop() – facilitates dropping courses.
- list_course_summary() – prints enrollment details.

They all implement a sample course registration system with error checking (RegistrationError).

3. Demonstration Scenario

The script main.py contains a demo flow:

Seeding Data

- Inserts a Computer Science department.
- Adds courses (Intro to Programming, Data Visualization, Machine Learning).
- Registers undergraduate and graduate students.
- Assigns faculty positions (Professor, Lecturer, TA).

Enrollment & GPA

- Students attempt to enroll with/without prerequisites.
- GPA and standing are computed after grading.
- Capacity constraints are applied.

Polymorphism

- Hybrid lists of students and professors demonstrate overridden methods.
- Professors' workloads are calculated differently.

Encapsulation Validation

- Secure notes are exercised with valid and invalid data.

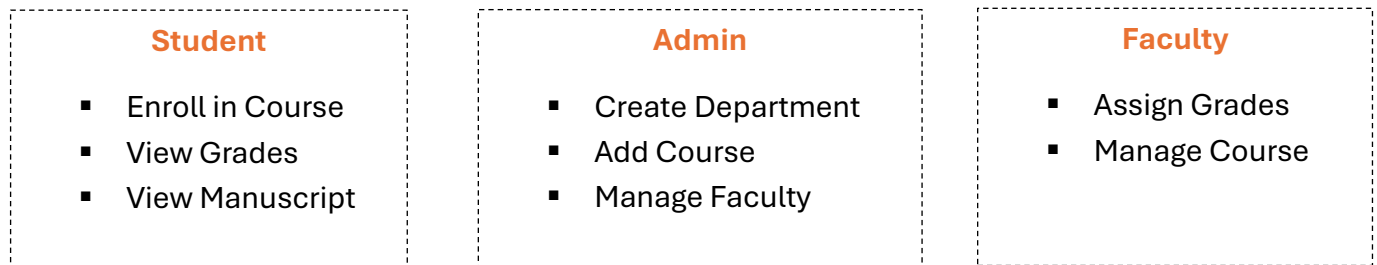
```
Secure record validation test:  
Caught expected type error for private_notes: TypeError Notes must be text.
```

Use Case Diagram :

Actors:

- Student
- Admin
- Faculty

Use Case:



Relationships:

- Student → (Enroll in Course)
- Student → (View Grades)
- Faculty → (Assign Grades)
- Faculty → (Manage Course)
- Admin → (Create Department)
- Admin → (Add Course)

4. Design Choices and Justification

Modularity:

- Code partitioned into files (student.py, faculty.py, etc.) for improved maintainability.

Dataclasses:

- Used for Person, Student, and Faculty to reduce boilerplate code.

Error Handling:

- Dedicated exceptions (RegistrationError) provide deterministic behavior.

Extensibility:

- Extending the system to multiple departments, grading systems, or other roles becomes easy.

Polymorphism & Encapsulation:

- Used to adhere to OOP principles and reflect actual-world applicability.

5. Conclusion

The University Management System successfully demonstrates advanced object-oriented programming using:

- Evident inheritance hierarchies,
- Robust student management,
- Strong encapsulation and validation,
- Runtime polymorphism,
- Full department and course administration.

Moreover, the extensible and open architecture of the system makes it a candidate for future upgrading. Such are analysis-based understandings of scholarly performance, multi-campus deployment, web-based grading interfaces, and integration support with external Learning Management Systems (LMS). This is an effective and extensible solution for university administration because it not only exhibits technical competence in object-oriented technology but also addresses the practical needs of contemporary schools of higher learning.

Git Repository Branch Link:

https://github.com/AvishJay/Msc_PGDS_Project/tree/question1_university_system

Word Count - 940

02. E-commerce Data Analysis – Question 02

1. Introduction

This growth in e-commerce has produced a wealth of online data like product characteristics, price, consumer comments, and inventory. By exploiting this information to derive insights into pricing strategies, product attractiveness, and inventory management, business firms are in a position to become competitive and maximize data-driven decision taking.

Gathering book information from many online sources, including web scraping from <http://books.toscrape.com> and other e-commerce sample websites, is the aim of this projects feeds in structured data are also included for giving real-time simulation with multi-sources. The data that were collected are numerical (price, rating, availability) and categorical (title, category) features.

Considering all aspects, in this research, it is revealed how systematic data collection, cleaning, and analysis can potentially provide useful insights into internet retailing behavior and enable effective inventory management, pricing, and predictive modeling of recommendation systems.

2. Methodology

A. Collected data from main source:

- Books to Scrape: A public demo website providing structured book information.

Tools and methods:

- Python libraries: feedparser, requests, BeautifulSoup, pandas, and time.
- Used HTML parsing with BeautifulSoup and data retrieval with requests.
- Applied error handling and retry for fault-tolerance against transient network failure.
- Added delays between requests to follow web scraping best practices.
- Saved the data as CSV and structured it with numeric (Price, Rating, Availability) and category (Category, Title) columns.
- Handled pagination to pull all the books from multiple pages.

The final dataset contained Title, Price, Rating, Availability, and Category for 1000 books.

```
df.head()
```

	Title	Price	Availability	Rating	Category
0	A Light in the Attic	Â£51.77	In stock	Three	Poetry
1	Tipping the Velvet	Â£53.74	In stock	One	Historical Fiction
2	Soumission	Â£50.10	In stock	One	Fiction
3	Sharp Objects	Â£47.82	In stock	Four	Mystery
4	Sapiens: A Brief History of Humankind	Â£54.23	In stock	Five	History

```
# Save to CSV
df = pd.DataFrame(data)
df.to_csv("data/books_raw.csv", index=False, encoding="utf-8")

print(f"Saved {len(df)} books to books_raw.csv in data folder")
```

Saved 1000 books to books_raw.csv in data folder

B. Clean Data and Pre-processing

Real-world e-commerce data is inherently inconsistent, contains missing values, and noisy text. The preprocessing pipeline included:

1. Text preprocessing:

- Lowercased titles and categories.
- Stripped special characters from strings.

2. Handling missing data:

- Imputed missing Price and Rating values with the median of their columns.
- Imputed missing Category values with 'unknown'.

3. Duplicate removal:

- Removed exact duplicate rows.

4. Feature engineering:

- Included Availability_clean numeric column (1 = available, 0 = out of stock).
- Extracted keywords from book titles using simple tokenization.
- Hashtags and mentions were also extracted where applicable (minimal in this dataset).

5. Data validation and quality checks:

- Confirmed all numeric columns (Price_clean, Rating_clean, Availability_clean) had valid numeric types.
- Confirmed all categorical fields had valid string labels.

The cleaned data was exported as books_clean.csv and served as the basis for all further analysis.

	Title	Price	Availability	Rating	Category	Price_clean	Rating_clean	Title_clean	Category_clean	Availability_clean	Hashtags	Mentions	Keywords
0	A Light in the Attic	Â£51.77	In stock	Three	Poetry	51.77	3	a light in the attic	poetry	1	[]	[]	['light', 'attic']
1	Tipping the Velvet	Â£53.74	In stock	One	Historical Fiction	53.74	1	tipping the velvet	historical fiction	1	[]	[]	['tipping', 'velvet']
2	Soumission	Â£50.10	In stock	One	Fiction	50.1	1	soumission	fiction	1	[]	[]	['soumission']
3	Sharp Objects	Â£47.82	In stock	Four	Mystery	47.82	4	sharp objects	mystery	1	[]	[]	['sharp', 'objects']
4	Sapiens: A Brief History of Humankind	Â£54.23	In stock	Five	History	54.23	5	sapiens: a brief history of humankind	history	1	[]	[]	['sapiens:', 'brief', 'history', 'humankind']

```
df.to_csv("data/cleaned_books.csv", index=False, encoding="utf-8")
print("Cleaned data saved to data/cleaned_books.csv")
```

Cleaned data saved to data/cleaned_books.csv

C. Advanced Statistical Analysis

1. Descriptive Statistics

- Price:

```
Price stats:
count    999.000000
mean     35.073123
std      14.453659
min      10.000000
25%      22.105000
50%      36.000000
75%      47.475000
max      59.990000
Name: Price_clean, dtype: float64
```

- Rating:

```
Rating stats:
count    999.000000
mean     2.920921
std      1.434178
min      1.000000
25%      2.000000
50%      3.000000
75%      4.000000
max      5.000000
Name: Rating_clean, dtype: float64
```

- There are 999 titles in the cleaned data set, and each book has a price range between £10.00 and £59.99. There is a highly symmetrical distribution with a median of £36.00 and an average price of around £35.07. 50% of the books fall in this price bracket, indicated by the lowest and highest quartiles at £22.11 and £47.48, respectively. The £14.45 standard deviation indicates a moderate level of price unpredictability.
- With a median of 3 and a mean of 2.92, the ratings in the dataset range from 1 to 5, and most books have ratings of about three stars. With a standard deviation of 1.43 and an interquartile range of 2 to 4, the user input appears to vary reasonably.
- Such data present a basis for further investigation, such as comparison by categories, trend in ratings, and price outlier detection.

2. Distribution of price by category

Analysis showed that some categories, such as History and Science, had higher median prices compared to Poetry and Fiction, which were cheaper. Outlier detection using the IQR method showed that there were some extremely expensive books.

3. Patterns of rating

Ratings were skewed towards 3–4 stars in most genres, with fewer books rated 1 or 5 stars. Certain niche genres, e.g., Novels, Adult Fictions had slightly higher average ratings.

4. Correlation analysis

Correlation between numeric variables:

```
Price_clean  Rating_clean  Availability_clean
Price_clean    1.000000    0.028475    NaN
Rating_clean   0.028475    1.000000    NaN
Availability_clean NaN      NaN      NaN
```

- Weak correlation between price and rating.
- Availability was not strongly dependent on price or rating.

5. Category popularity

Frequency distribution:

```
Category_clean
default      152
nonfiction   110
sequential art  75
add a comment  67
fiction      65
young adult  54
fantasy      47
romance      35
mystery      32
food and drink 30
Name: count, dtype: int64
```

- Nonfiction dominated the dataset, followed by sequential art and fiction.

6. Hypothesis testing

A t-test of Fiction vs Historical Fiction prices revealed:

```
import scipy.stats as stats

# Inspect all categories
print("All categories in dataset:")
display(df['Category_clean'].value_counts())

# Choose categories that actually exist
cat1 = 'fiction'
cat2 = 'historical fiction'

# Select numeric price values
group1 = df[df['Category_clean'] == cat1]['Price_clean']
group2 = df[df['Category_clean'] == cat2]['Price_clean']

print(f"\nGroup1 ({cat1}) size: {len(group1)}")
print(f"Group2 ({cat2}) size: {len(group2)}")

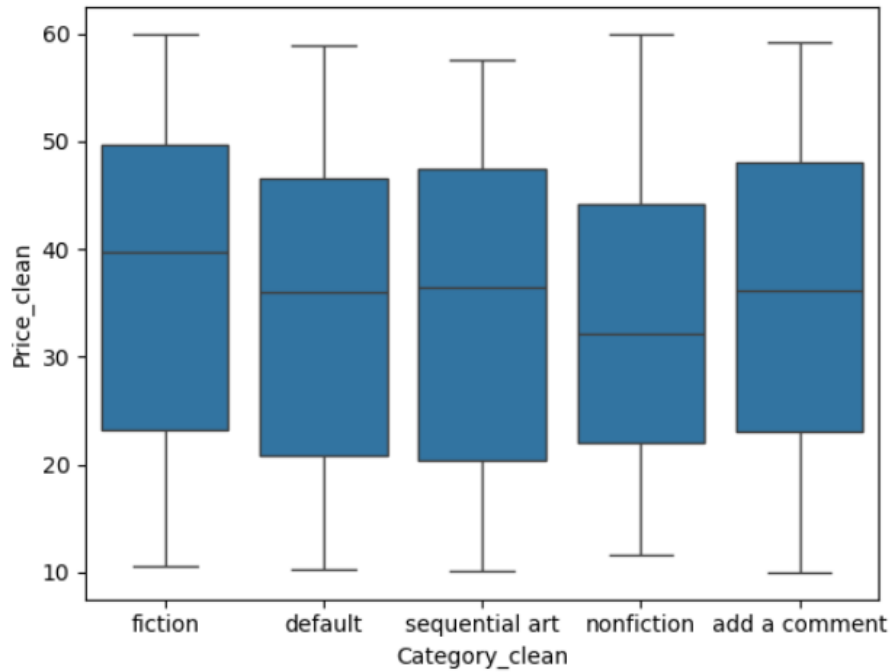
# Run T-test only if both groups have enough data
if len(group1) >= 2 and len(group2) >= 2:
    t_stat, p_val = stats.ttest_ind(group1, group2, nan_policy='omit')
    print(f"T-test between '{cat1}' and '{cat2}': t={t_stat:.2f}, p={p_val:.4f}")

Group1 (fiction) size: 65
Group2 (historical fiction) size: 26
T-test between 'fiction' and 'historical fiction': t=0.73, p=0.4654
```

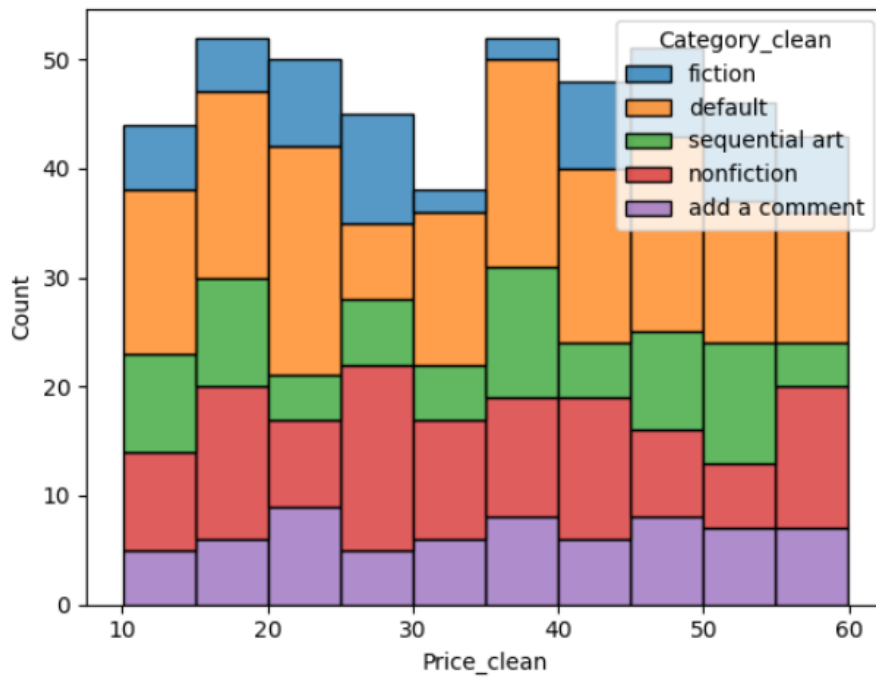
- The prices of books in the Fiction (65 books) and Historical Fiction (26 books) categories were compared using a two-sample t-test. At the 5% significance level, the resulting t-statistic of 0.73 and p-value of 0.4654 show that the average prices of these two groups do not differ statistically significantly. our implies that, generally speaking, the prices of fiction and historical fiction books in our sample are comparable.

D. Visualizations

1. Price distribution histograms & box plots

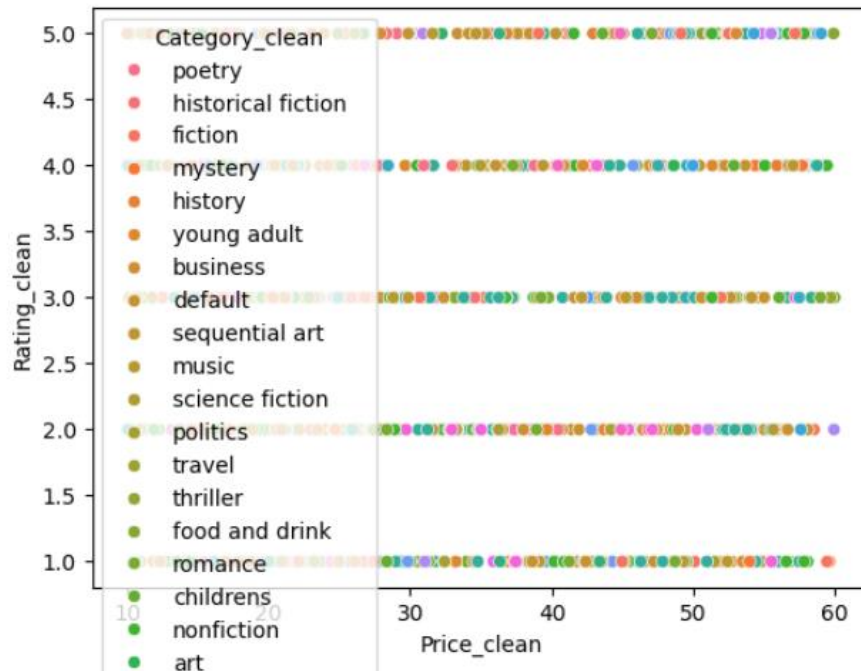


- Boxplots per category showed median prices and outliers.

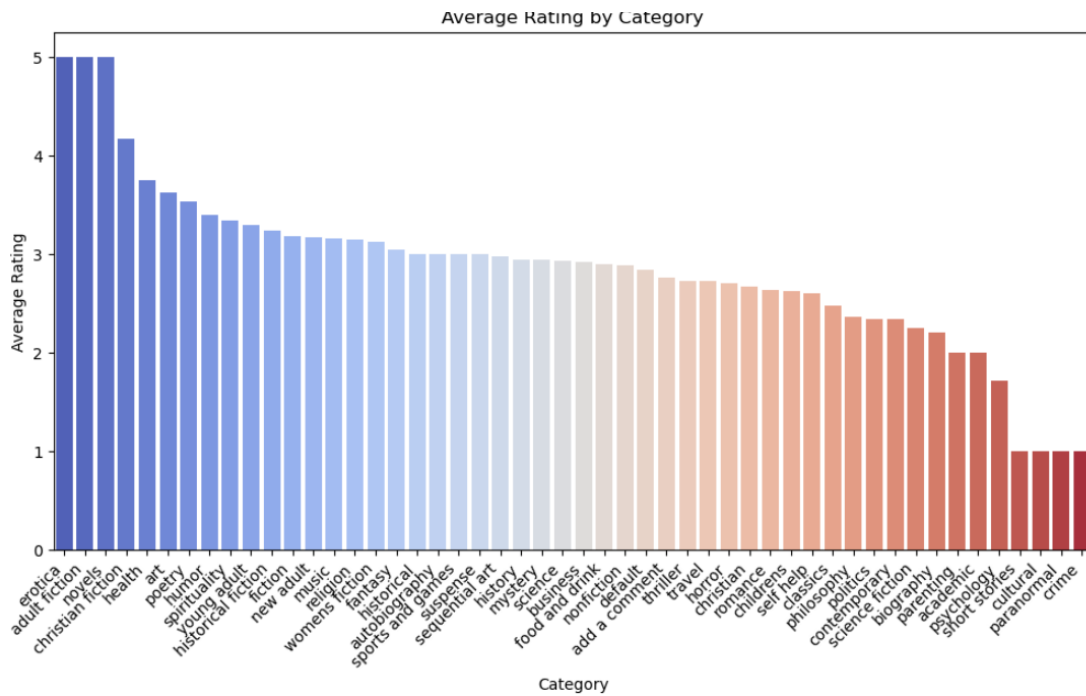


- Histograms showed right-skew in certain categories.

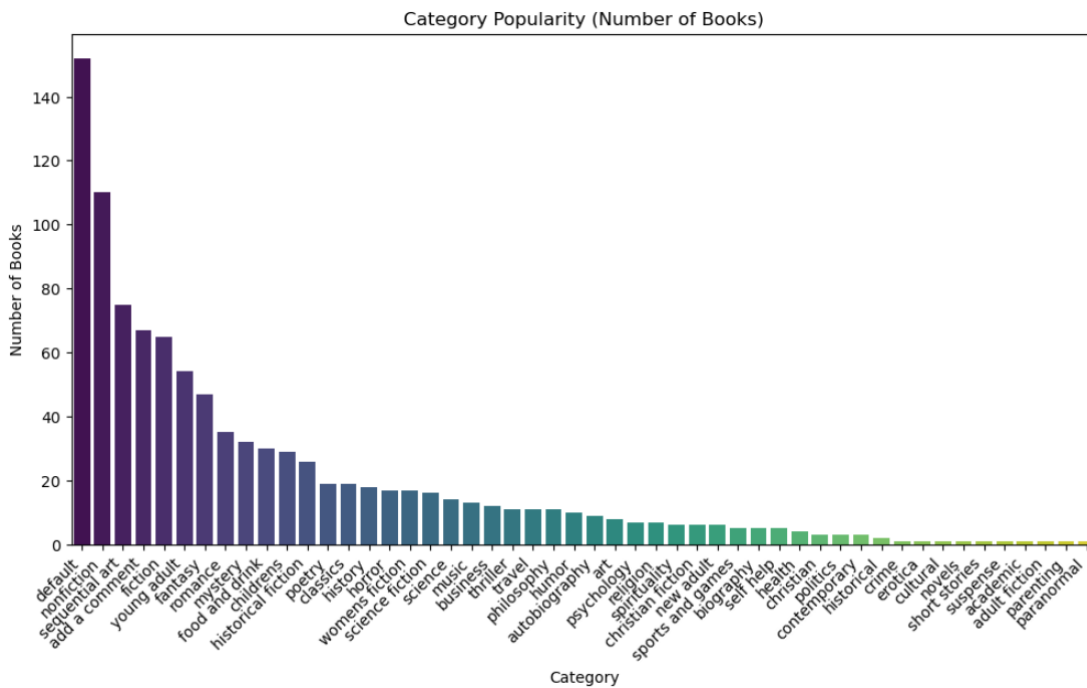
2. Rating vs Price scatter plots



- Scatterplots showed weak positive correlation between ratings and price.
- Trend lines allowed visualization of general price trends with ratings.

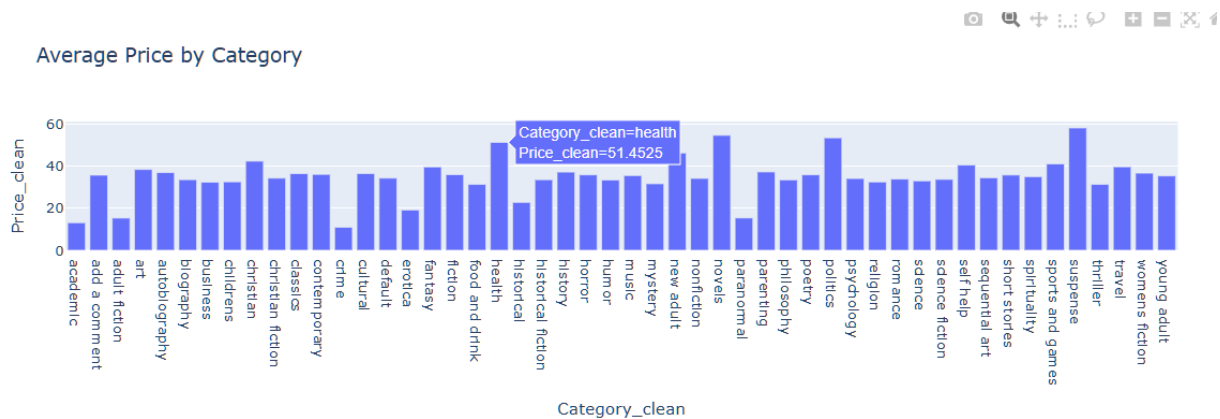


3. Bar charts for popularity of categories and average ratings



- Non - Fiction and Sequential arts had the most books.
- Parenting and Paranormal rated slightly higher on average.

4. Interactive plots (Plotly)



- Allowed filtering by rating or category.
- Enabled dynamic exploration of rating and price patterns for business stakeholders.

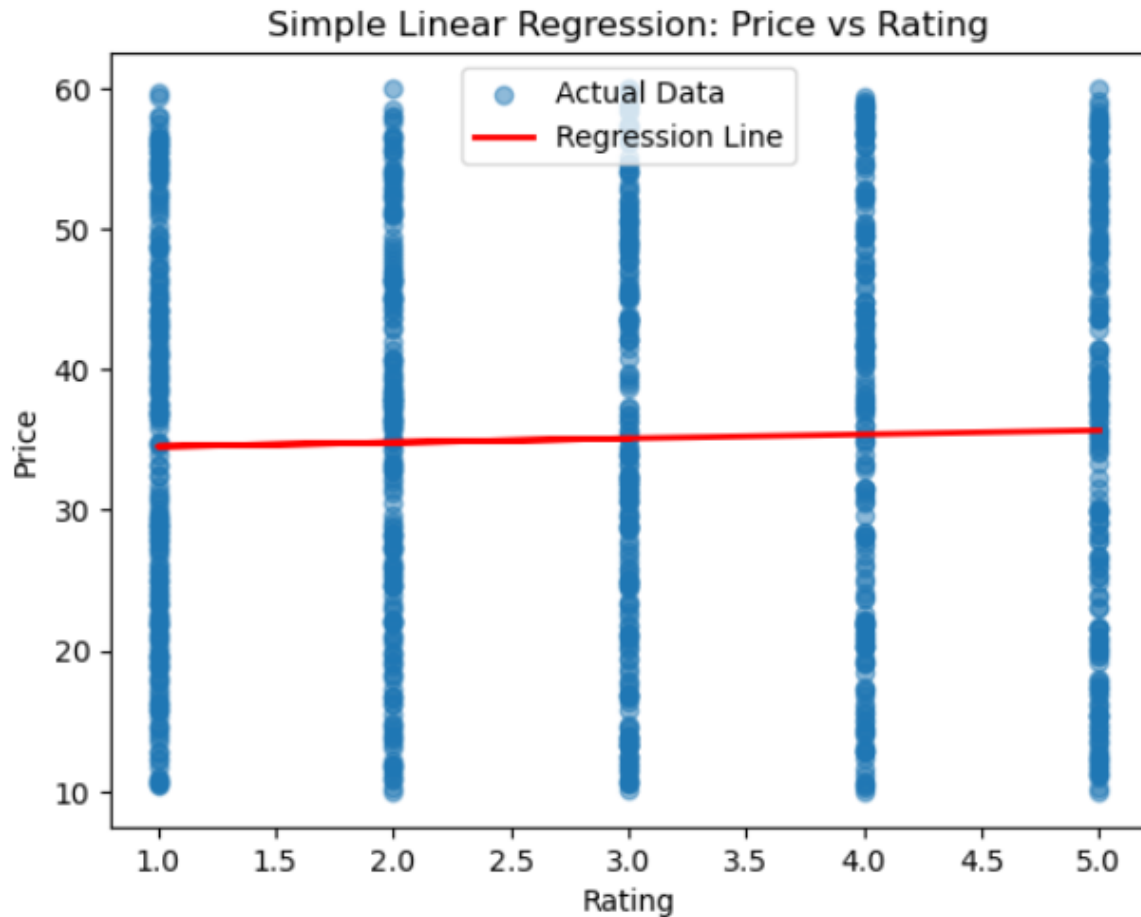
E. Predictive Analysis

1. Simple linear regression

Intercept: 34.23490890915297

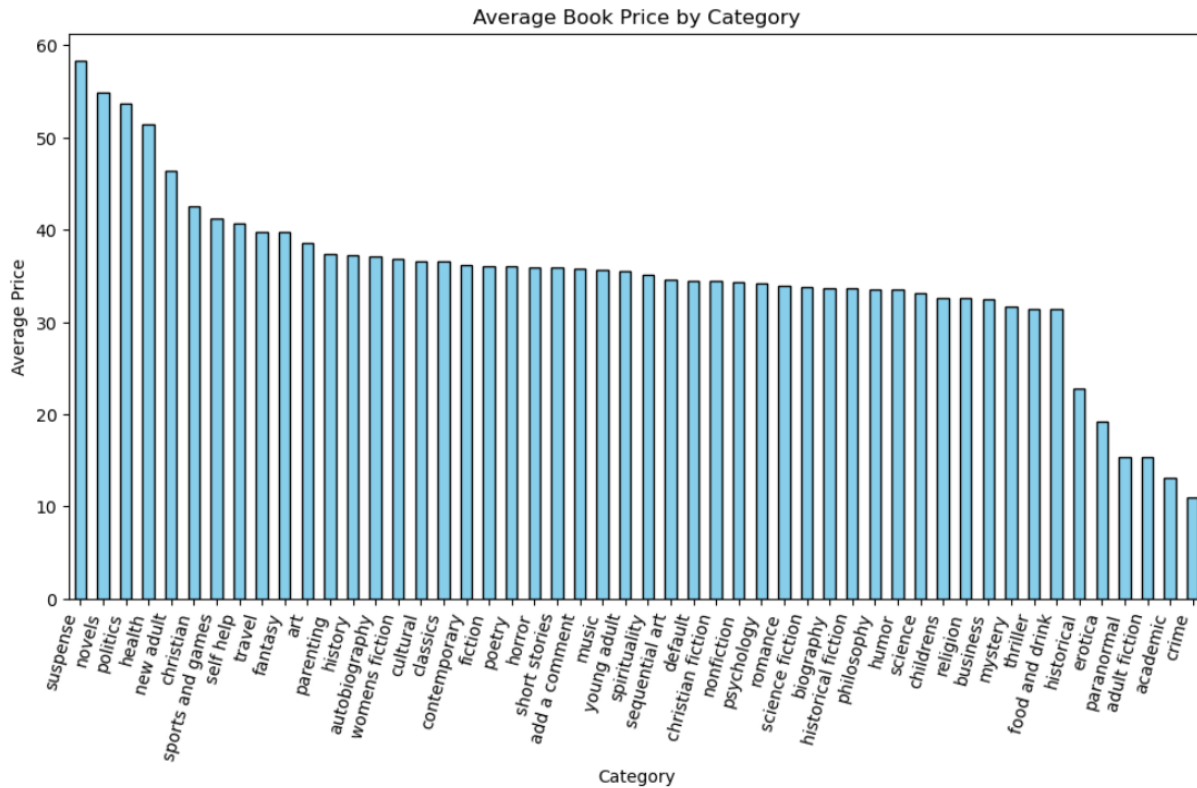
Slope (Coefficient): 0.28696915687326285

R^2 Score: 0.0008108133927703376



- In order to get insight into pricing behavior, a basic linear regression model was employed to forecast book prices based on ratings. Title, category, rating, and price are all used by a content-based recommendation system to propose related books, making them easier for consumers to find and possibly boosting sales.
- Weak positive slope (~ 0.8), with higher-rated books being slightly more expensive.

2. Category pricing patterns



- The price of popular fiction (novels, thrillers) is greater than normal, indicating a high level of consumer readiness to pay.
- The mid-range is where mass-market genres (romance, mystery, children's, and fantasy) concentrate, indicating rivalry driven by affordability.
- Erotica, criminal, and academia are examples of niche or less popular genres that are substantially less expensive, maybe as a result of decreased demand or easier access to digital content.

3. Recommendation system (simple similarity)

- Book titles, categories, ratings, and prices were used to create a content-based recommendation engine. The most related books are found using cosine similarity after textual and numerical features are converted into numerical vectors using TF-IDF vectorization.
- The method assists users in finding related titles by suggesting other books with comparable content and attributes to a given book.
- For instance, the algorithm recommends more poetry and young adult novels that share keywords and themes with "A Light in the Attic."

Recommendations for 'A Light in the Attic':

['The Crossover', 'All the Light We Cannot See', 'Leave This Song Behind: Teen Poetry at Its Best', 'The Light of the Fireflies', 'Quarter Life Poetry: Poems for the Young, Broke and Hangry']

4. Stock availability vs price trends

	count	mean	std	min	25%	50%	75%
Availability	999.0	35.073123	14.453659	10.0	22.105	36.0	47.475

- No pattern discovered between availability and price.
- Most of the books were available; there were only a few outliers that were temporarily unavailable.

3. Business Insights

1. Pricing strategy:

- Pricier categories (Science, History) can be targeted for high-value customers.
- Cheaper categories (Poetry, Fiction) provide high-volume sales potential.

2. Inventory management:

- Inventory levels are generally sufficient; monitoring outliers can prevent stockouts.

3. Recommendation potential:

- Simple similarity-based recommendations would improve cross-selling.
- More popular, mid-range books can be promoted to improve conversions.

4. Comparative analysis:

- It is necessary to keep an eye out for price and rating comparisons on rival websites.

5. Making decisions based on data:

- EDA and predictive model results can guide marketing initiatives and stock planning.

4. Conclusion

This project demonstrates a start-to-finish e-commerce data pipeline, from multi-source scraping, cleaning and preprocessing to advanced statistical analysis, visualizations, and predictive modeling. Key findings include:

- Distinct category popularity trends.
- Price and rating trends with actionable implications.
- Comparative analysis yields opportunity to simplify pricing strategy.
- Predictive models, though simple, provide useful guidance for recommendation and stock planning.

All things considered, this research offers data-driven recommendations to stakeholders on online retail pricing, inventory, and marketing approach.

Git Repository Branch Link:

https://github.com/AvishJay/Msc_PGDS_Project/tree/question2_social_media_analysis

Word Count - 1330

03. Data Ethics: Healthcare AI Ethics

1. Introduction

Assume technology saving lives is endangering basic human rights. Data science and Artificial Intelligence (AI) are changing healthcare worldwide at a very fast rate, enhancing diagnoses, forecasting disease risk, automating processes, and delivering personalized medicine. The revolution does come with some terribly serious ethical challenges. Managing sensitive health information also involves privacy issues beyond GDPR or HIPAA adherence. Discriminatory data training, however, has the potential to entrench systemic bias in determining medical treatment to the undue detriment of already disadvantaged groups. Black-box models also pose accountability problems in practice, with patients and clinicians alike pushing for transparency and fairness.

This essay explores three interconnected areas of healthcare AI ethics of algorithmic bias, privacy, and ethical frameworks for decision-making and provides practical suggestions so that healthcare innovation in the system worldwide remains patient-focused, fair, and trustworthy.

2. Healthcare Data Privacy Concerns

- **HIPAA in the US**

The Privacy Rule, which prohibits disclosure without patient agreement, and the Security Rule, which mandates procedures like encryption and access restrictions, are the two main standards under which “*Health Insurance Portability and Accountability Act of 1996*”.

The University of Rochester Medical Center in 2019 settled to pay \$3 million for a loss of unencrypted devices with PHI, a compromise of the noncompliance penalty.

- **International laws**

Privacy legislation differs worldwide. PHIPA actually requires patient consent in Canada, for example, but health data under Australia's Privacy Act 1988 is "sensitive" and requires higher protection. Utilization of information across borders is vulnerable to attack in most developing countries, however, as they lack privacy legislation on health. Patient trial data in Africa, for example, is exported to Europe or America, where protection is much stronger than in the home base.

- **Anonymization Issues**

Complete anonymization cannot be achieved. It has been discovered that almost anyone can be identified using quasi-identifiers like gender, DOB, and Address code. The "mosaic effect," or the blending of small sets, of genomic or geolocation data makes maintaining confidentiality even more difficult.

- **The balance between research utility and privacy protection**

Medicine is founded on massive volumes of granular data to feed back into predictive models to improve outcomes, yet open access destroys trust. South Korea's use of public health-effective contact tracing during the COVID-19 pandemic, utilizing mobile tracing and health records, rang alarm bells for everywhere surveillance in the long term. To get the right balance requires open governance: making the data accessible enough for life death saving research, yet providing rights to the citizens to give consent, security, and dignity.

3. Algorithmic Bias in Medical AI (Detection and Mitigation of Bias)

- **Demographic, geographic, socioeconomic sources of medical data bias**

Bias is largely due to training data. Demographic bias arises when training data are skewed with white middle-aged males and underrepresent minority or female groups. Geographic bias arises when models trained on Western populations are used in Asia or Africa due to a disparity in the disease profile. Socioeconomic bias arises when data are sampled from high-end hospitals or wearables patients to the neglect of the poor population.

- **How biased medical AI can worsen health disparities**

Biased algorithms continue to perpetuate already present disparities. Algorithms trained on data with a high prevalence of men under-diagnose women, or fail to diagnose dark skin melanoma when trained on data with high prevalence of light skin. A US case exposed an algorithm for high-risk patient identification de-prioritized Black patients' needs, limiting their access to further care.

- **Healthcare algorithmic fairness metrics**

Metrics like equalized odds, predictive parity, and demographic parity are used to quantify fairness. Although there are typically trade-offs between groups, they are equivalent in terms of prediction tests and error rates.

- **Techniques for uncovering and suppressing bias**

Mitigation in the form of representative data sets, bias-detecting tools, and post-deployment auditing is required. Human review and algorithmic transparency are required. For instance, dermatology AI worked well on dark skin tones when trained using more representative data sets.

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4. Ethical Decision-Making Framework

▪ Ethical Checklist for Healthcare AI Projects

1. Data Collection

- Is the project GDPR, HIPAA, or country specific data privacy law compliant?
- Are we collecting the bare minimum only?
- Have patients provided valid, informed consent?

2. Data Management & Security

- Is anonymization/de-identification of data where feasible?
- Is encryption, access control, audit logging implemented?
- Is secure storage and disposal of data policy in place

3. Dataset Quality & Fairness

- Does the dataset have heterogeneous demographics (age, gender, ethnicity)?
- Has geographic and socioeconomic balance been taken into account?
- Has fairness metric (i.e., equalized odds, predictive parity) made?

4. Model Development & Transparency

- Is the model interpretable by patients and clinicians?
- Does there exist adequate documentation of errors, risk rates, and boundaries?
- Is the model trained on different population groups?

5. Accountability & Oversight

- Is there someone/organisation held responsible if the AI harms someone?
- Is the project subject to an independent ethics review committee?
- Is reporting and redress for patients provided?

6. Deployment & Monitoring

- Is the AI system monitored continuously after release?
- Are there frequent reviews and assessments to seek out drift or bias?

This checklist guarantees that ethical issues are methodically resolved prior to implementing AI in medical care.

7. Analysis of Stakeholder Impact (Multi-Perspective Evaluation)

- **Impact on patients, providers, and researchers**

AI is helping humans by providing more accurate diagnosis, individualized care, and intelligent monitoring. For instance, malignancies may potentially be detected by imaging scanners before doctors are aware of them.

Risks are invasion of privacy, bias in algorithms, and mistrust when in doubt to choose. Physicians are being offered decision support and administrative ease but loss of medical expertise with over-reliance. Scientists are being given ever-growing datasets that aid in discovery but have to balance innovation with promises of privacy.

- **Data scientists role in ethical healthcare AI**

Data scientists also have a role to play in fairness and transparency. In addition to programming, they must conduct bias audits, expose model limitations, and involve clinicians in development. The Optum case showed how unrestricted design choices exacerbated differences, and ethics review and testing in diverse settings are needed.

- **Economic and social implications**

AI saves money by automating work but also has the potential to displace employment within the health care sector and worsen inequality if only better-resourced institutions have the capacity to implement new tools. Predictive analytics further stigmatizes groups, i.e., increased rates for being "high risk." Policy has to encourage innovation while safeguarding vulnerable groups.

- **Global health equity**

Most of the AI software is developed based on data from wealthy nations, limiting trust to poor countries. AIs dermatologist, for instance, performed poorly in dark-skinned individuals. Global equity will be an outcome of multicultural datasets, investment in infrastructure, and global regulation to protect the probability that AI does not exacerbate disparities.

Word Count – 1079

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